

Solution Requirements

Date	1 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Team members and administrators log in to the Flask web application using a username and password; passwords are stored using secure hashing before authentication is granted.	Medium
2	Authorization levels	Two roles are supported: Admin (manages datasets, retrains/updates the model, views all predictions) and Standard User / Meteorologist (enters weather data and views flood predictions only).	Medium
3	External interfaces	The system connects to the Kaggle-sourced flood prediction dataset for training, and is designed for deployment on IBM Cloud so the web application can be accessed from any location.	High
4	Transactions processing	User-submitted rainfall, cloud visibility and seasonal rainfall values are validated, scaled using the saved scaler, and passed to the trained XGBoost model to generate a prediction in real time.	High
5	Reporting	The system displays a Flood Chance or No Flood Chance result page for each submission and allows authorities to review recent predictions per region.	Medium
6	Business rules, etc.	A prediction is only generated once all required weather inputs (annual rainfall, cloud visibility, seasonal rainfall) are provided and pass validation; incomplete submissions are rejected with an error message.	High
7	Compliance to laws or regulations	User and regional data handled by the system follows standard data-protection practices; no personally identifiable information beyond login credentials is stored.	Low

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
8	Other	The best-performing trained model and scaler are saved as floods.save and loaded automatically by the Flask application on startup for consistent predictions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction pipeline (input validation, scaling, and model inference) must return a flood risk result quickly enough for real-time decision-making during monsoon monitoring.	Prediction result returned in under 3 seconds
2	Scalability	The Flask application and underlying model should handle prediction requests from multiple regions and users at once, especially during active flood monitoring periods.	Supports at least 100 concurrent prediction requests
3	Security & Data Privacy	Login credentials and any regional data submitted through the app must be protected from unauthorized access.	Passwords hashed; data transmitted over HTTPS
4	Reliability & Availability	Since the tool supports time-critical flood warnings, the web application must remain available and resilient, particularly during monsoon season.	99.5% uptime per month
5	Usability & Accessibility	The Home, Prediction Input, and Result pages must be simple enough for non-technical disaster management staff to use with minimal training.	New users complete a prediction within 2 minutes without assistance
6	Other	The model and codebase should be maintainable so the team can retrain on updated datasets and redeploy on IBM Cloud without major rework.	Model retraining and redeployment completed within 1 day